

DS5899: Special Topics in Data Science

Leveraging NLP in Asset Management

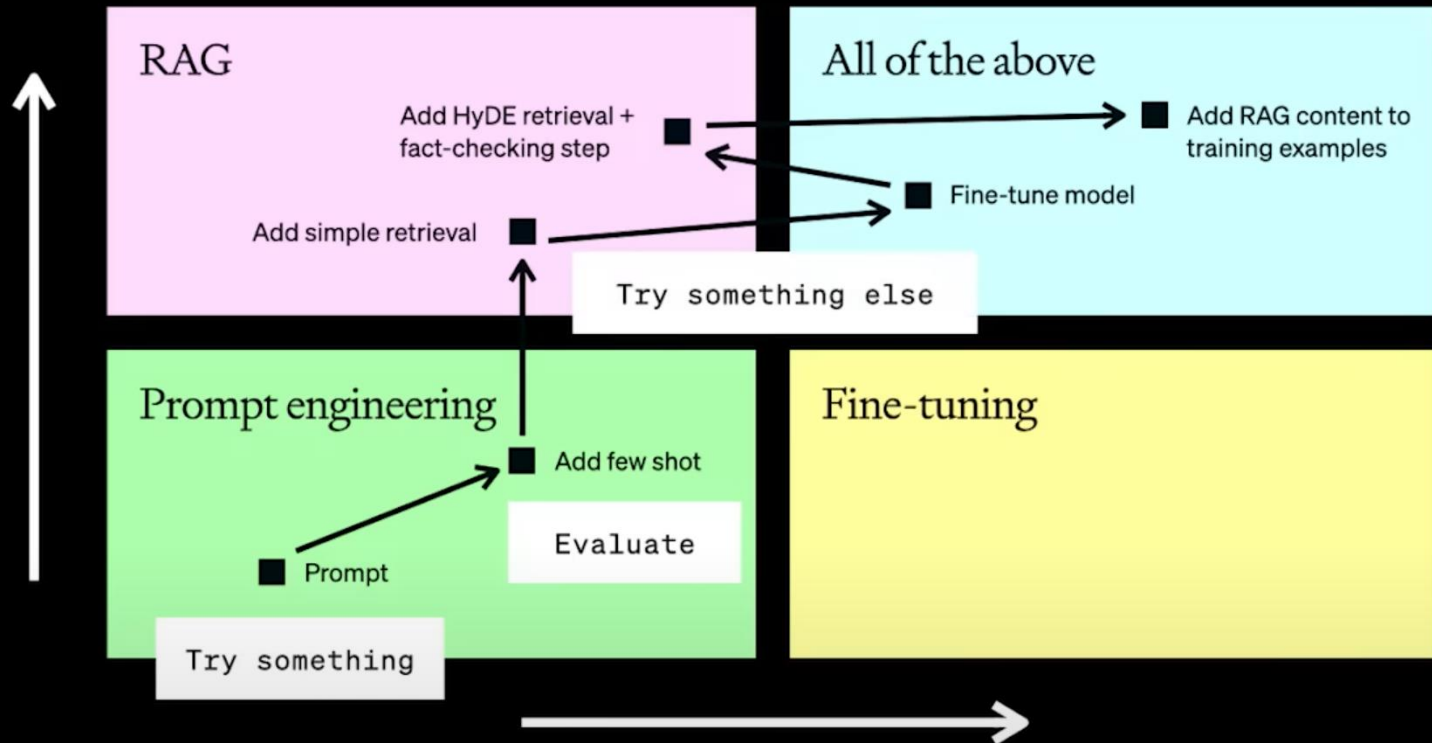
Week 8: Prompting

Instructor: Che Guan, Ph.D.

The optimization flow

Context optimization

What the model needs to know



LLM optimization

How the model needs to act

Part 1:

Prompt Engineering

LLMs Are Getting Longer Contexts

Proprietary Models	Context Window	Notable Capability	Open Source Models	Context Window	Best Use Case
Google Gemini 3 Pro	1M - 2M tokens	The gold standard for multi-modal context (video/audio).	Meta Llama 4 Scout	10M tokens	Currently holds the record; designed to replace RAG.
Claude 4.5 Sonnet	1M tokens	Exceptional at "agentic" tasks like coding across entire repos.	Mistral Large 3	256K tokens	Massive 675B MoE model; very efficient for its size.
OpenAI GPT-5.2	400K tokens	Focused on high-reasoning accuracy over raw length.	Qwen 3-Coder	256K - 1M*	Specialized for repository-scale code understanding.
Claude 4.6 Opus	1M tokens	Features "Infinite Chats" to mitigate context errors.	DeepSeek-R1	164K tokens	A reasoning-first model with high "needle-in-a-haystack" recall.
Grok 4.1 Fast	2M tokens	Optimized for real-time data ingestion from X (Twitter).	Kimi K2.5	256K tokens	Popular for high-speed document reasoning.

Lost in the Middle: How Language Models Use Long Contexts

- Changing the location of relevant information within the language model's input context results in a U-shaped performance curve.

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1](Title: Asian Americans in science and technology) Prize in physics for discovery of the subatomic particle J/ψ . Subrahmanyam Chandrasekhar shared...

Document [2](Title: List of Nobel laureates in Physics) The first Nobel Prize in Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...

Document [3](Title: Scientist) and pursued through a unique method, was essentially in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics

Answer:

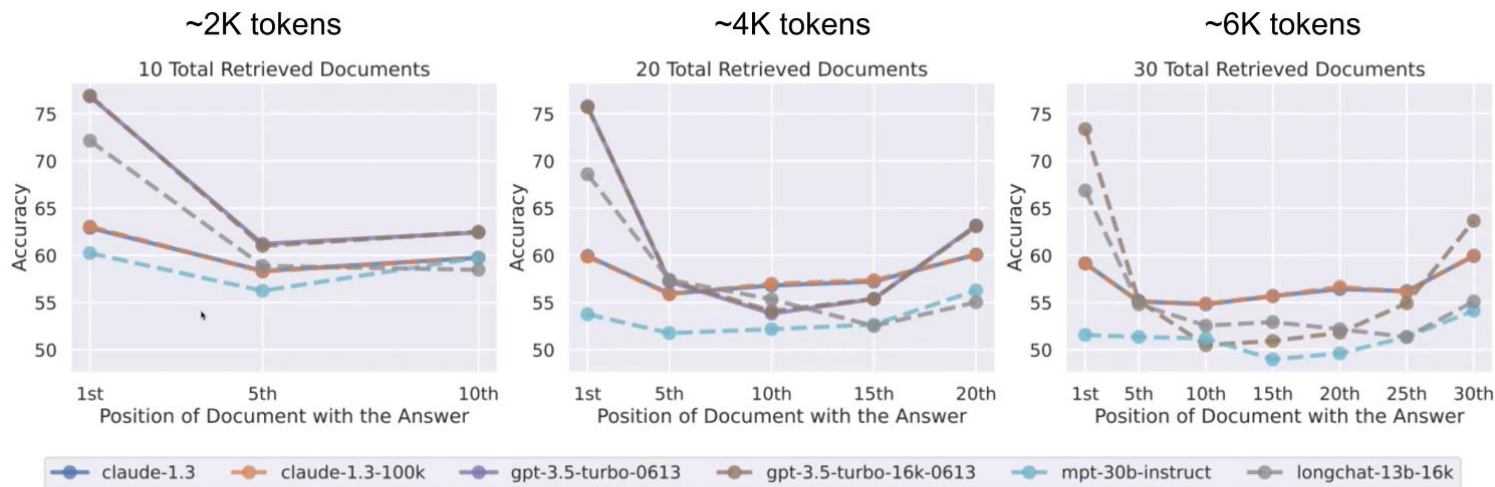
Desired Answer

Wilhelm Conrad Röntgen

Figure 2: Example of the multi-document question answering task, with an input context and the desired model answer. The document containing the answer is bolded within the input context here for clarity.

A U-shaped Performance Curve

- Models are particularly good at using information at the beginning (primacy bias) or the end (recency bias) of their input context. However, their performance tends to decline when they need to recall and apply information located in the middle of the input context.

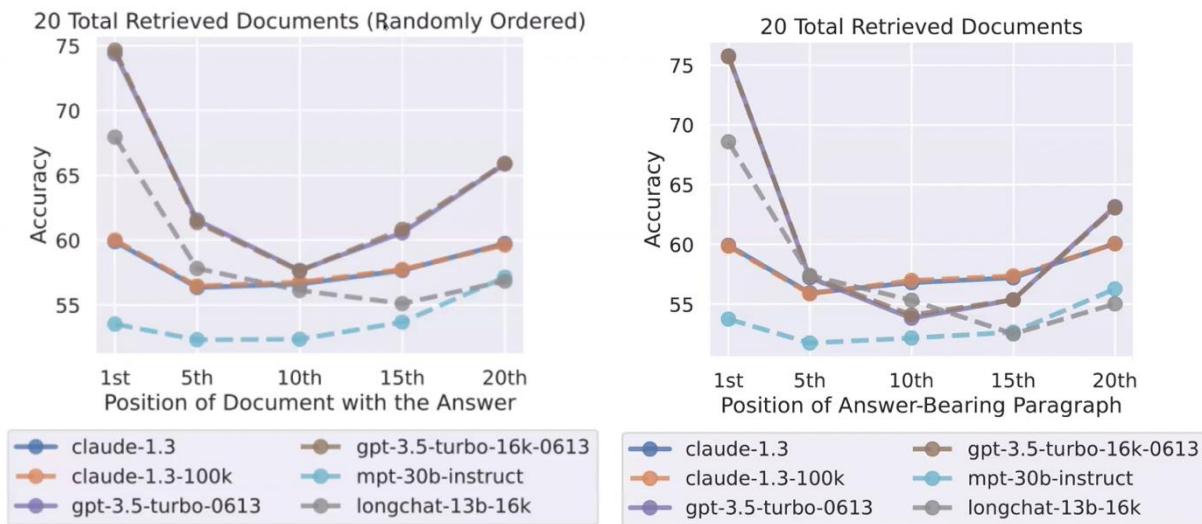


Best Closed-Book performance: GPT-3.5-Turbo, ~56%

Best Oracle (only feed in relevant doc) performance: GPT-3.5-Turbo, ~88.5%

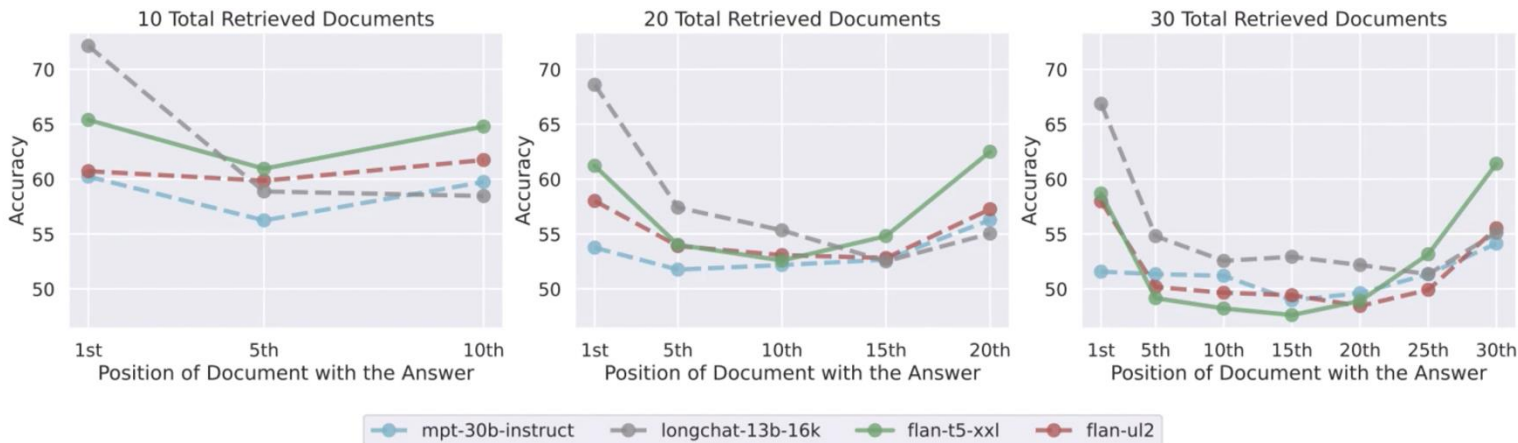
Is It Due to the Prompt Mention “Search Results”?

- To see if model could be "forced" to pay equal attention to the entire context, instructions are changed:
 - **Original Prompt:** Simply asked the model to use the search results.
 - **Modified Prompt:** Explicitly told the model that the results were **ordered randomly** and that some might be **irrelevant**.



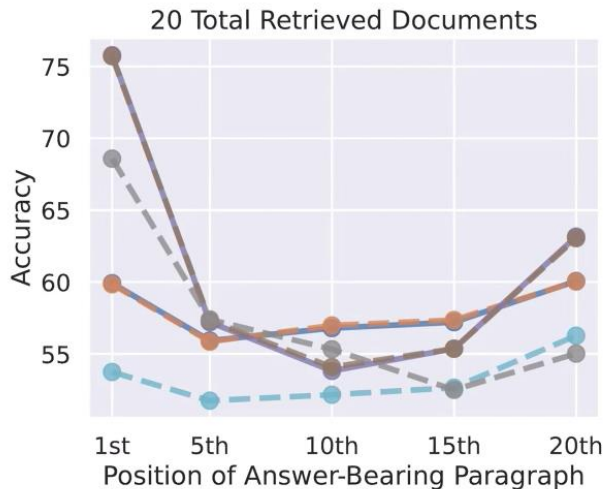
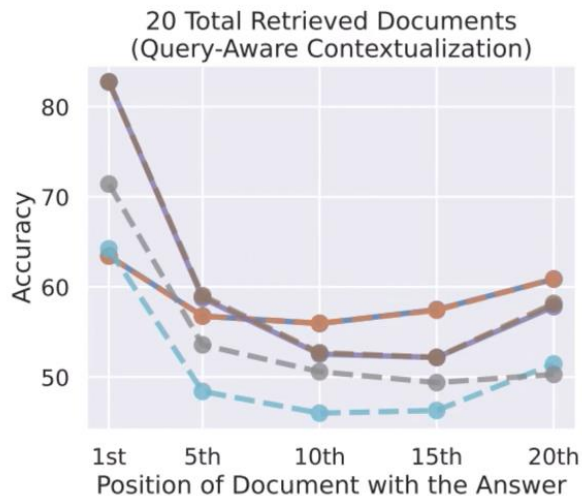
Is It Due to the Effect of Decoder Model Architecture?

- When the total amount of information provided to the model is within **2048 tokens**, the model is operating within its "comfort zone".
- Once a prompt has longer than 2048 tokens, the model's performance begins to degrade in a very specific pattern known as the U-shaped curve.
- 'Encoder-Decoder' models (like Flan-UL2 and Flan-T5-XXL) actually perform slightly better than "Decoder-only" models on longer sequences, but they still eventually succumb to the U-shaped degradation.



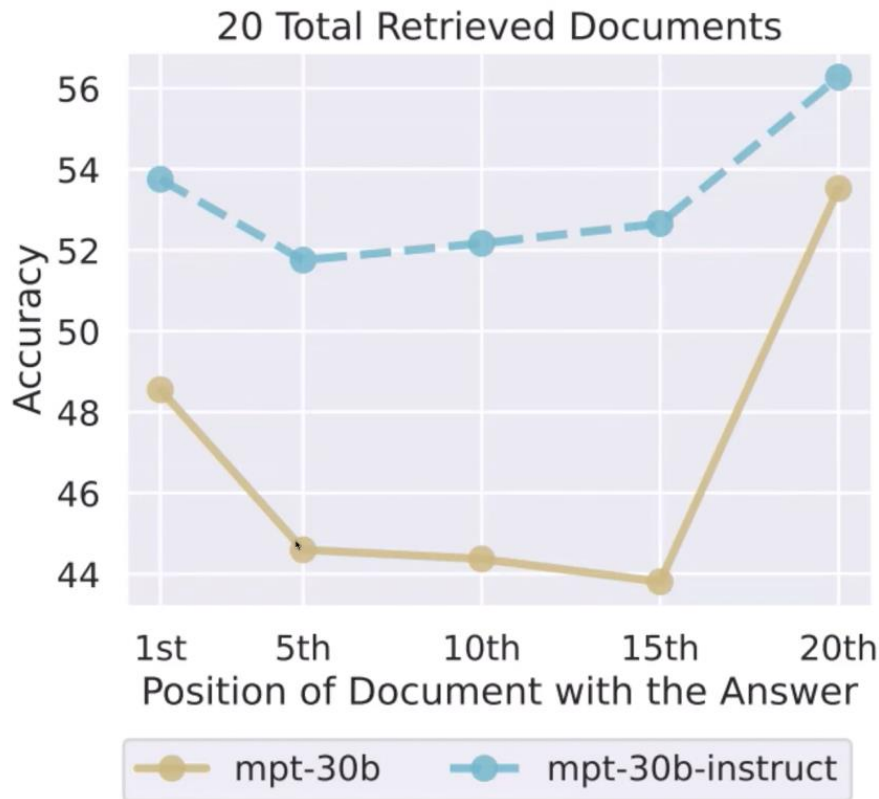
Does the Placement of the Query at the End of the Context Have an Impact on the Result?

- Since decoder-only models don't see the question until the very end of the context, their representations of document tokens are query-independent
- What if we put the query before *and* after the documents?



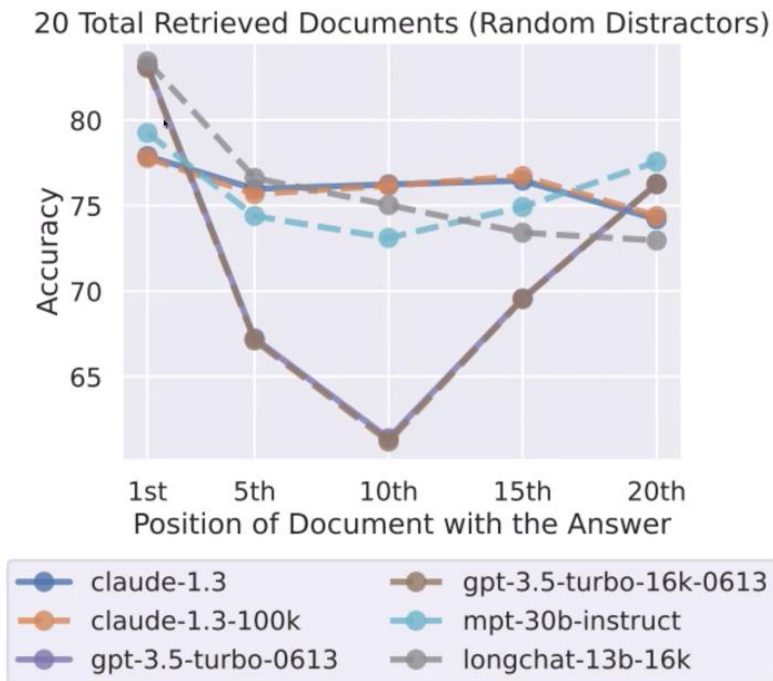
Is It Due to Effect of Instruction Fine-tuning?

- Intuition: instruction fine-tuning generally places instruction at very beginning or very end, which may cause this U-shaped perf.
- Surprisingly, that doesn't seem to be in the case.



Is It Because LMs Get Distracted by Retrieved Documents?

- What happens if you use random distractors instead of retrieved distractors?
- Surprisingly, model performance still sees a sizable degradation.
- Random Distractors: documents chosen completely at random from a large dataset.
- Retrieved Distractors: documents topically related to the user's question but do not actually contain the answer.



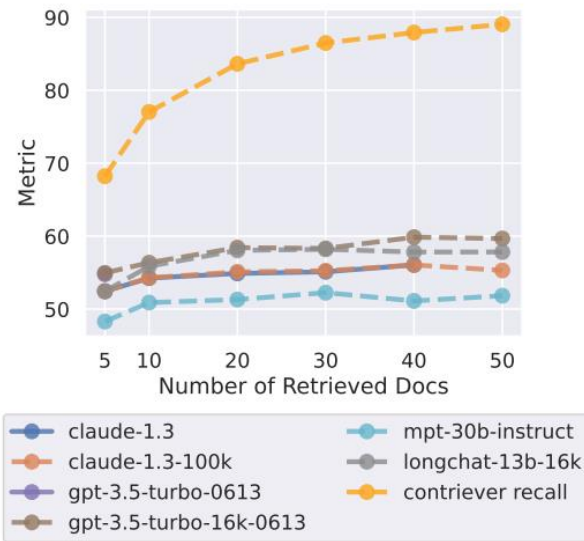
In Summary

- **Is this due to the prompt mentioning “search results”?**
 - No, modified prompt still shows degradation
- **Is this specific to decoder-only LMs?**
 - Encoder-decoder slightly better but still degrades on longer contexts.
- **Is this because we put the query only at the end of the context?**
 - No, putting query at start and end still results in degradation
- **Is this a byproduct of instruction-tuning?**
 - No, non-instruction tuned models also show these trends.
- **Is this because LMs get distracted by retrieved documents (hard negatives*)?**
 - No, performance degrades also on random negatives

* In information retrieval, a "hard negative" refers to a document that is considered irrelevant to a given query but is very similar in content or features to a relevant document, making it difficult for a retrieval model to accurately distinguish between the two.

Is More Context Always Better?

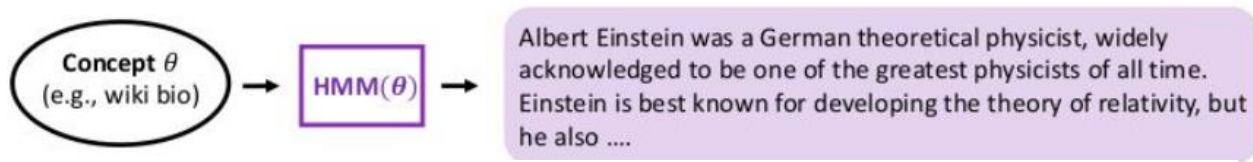
- As the number of retrieved documents increases, model performance saturates long before retriever recall, indicating that the models have difficulty making use of the extra retrieved documents.



***Contriever recall** can be interpreted as the probability that a relevant document will be retrieved by the model.

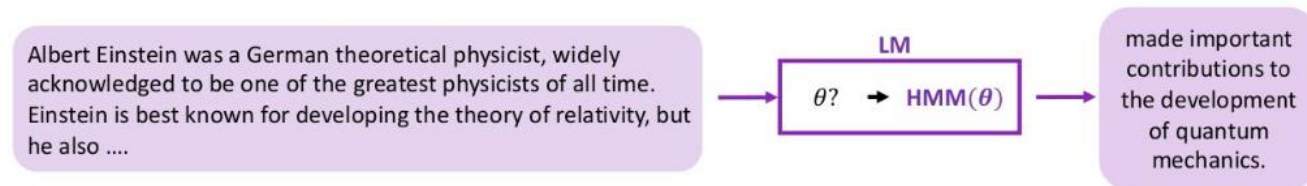
Pretraining distribution: mixture of HMMs (Hidden Markov Models)

Concept θ encodes transitions in HMM



Intuition

Language model **implicitly** infers latent concept θ

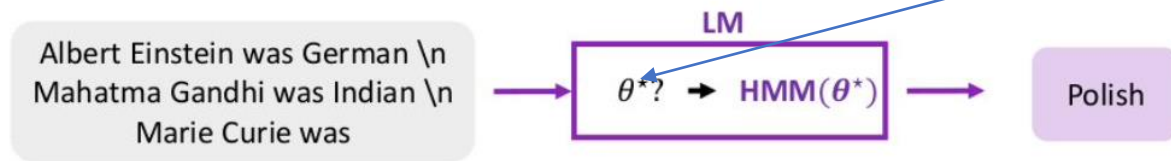


Bayesian inference view of in-context learning

Intuition

Can the language model infer concept θ^* ?

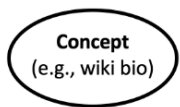
all the components of the prompt (input distribution, the output space and format) are providing “evidence” to enable the model to better infer (locate) concepts that are learned during pretraining.



Difficulty: condition on samples from prompt distribution, not training distribution!

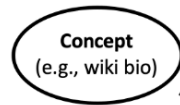
Bayesian inference view of in-context learning, Continued

1. Pretraining documents are conditioned on a **latent concept** (e.g., biographical text)



Albert Einstein was a German theoretical physicist, widely acknowledged to be one of the greatest physicists of all time. Einstein is best known for developing the theory of relativity, but he also

2. Create **independent examples** from a **shared concept**. If we focus on full names, wiki bios tend to relate them to nationalities.



Input (x)	Output (y)	Delimiter
Albert Einstein was	German	\n
Mahatma Gandhi was	Indian	\n
Marie Curie was	?	...brilliant? ...Polish?

3. **Concatenate examples into a prompt** and predict next word(s). **Language model (LM)** implicitly **infers the shared concept** across examples despite the unnatural concatenation

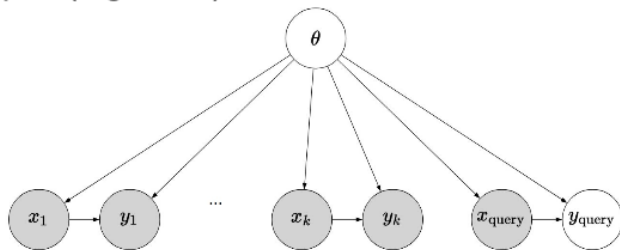


Figure 1: In-context learning can emerge from modeling long-range coherence in the pretraining data. During pretraining, the language model (LM) implicitly learns to infer a latent concept (e.g., wiki bios, which typically transition between name (Albert Einstein) → nationality (German) → occupation (physicist) → ...) shared across sentences in a document. Although prompts are unnatural sequences that concatenate independent examples, in-context learning occurs if the LM can still infer the shared concept across examples to do the task (name → nationality, which is part of wiki bios).

In-context Learning as Implicit Bayesian Inference

Bayesian inference

Posit a latent concept θ (e.g., task)

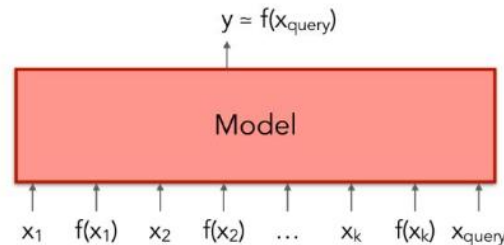


$$\underbrace{p(y_{\text{query}} \mid x_1, y_1, \dots, x_k, y_k, x_{\text{query}})}_{\text{posterior predictive}} = \int p(y_{\text{query}} \mid x_{\text{query}}, \theta) \underbrace{p(\theta \mid x_1, y_1, \dots, x_k, y_k)}_{\text{posterior}} d\theta$$

Transformer directly fits the posterior predictive distribution!

$$p(\text{output} \mid \text{prompt}) = \int_{\text{concept}} p(\text{output} \mid \text{concept}, \text{prompt}) p(\text{concept} \mid \text{prompt}) d(\text{concept})$$

- Ideally, $p(\text{concept} \mid \text{prompt})$ concentrates on the prompt concept with more examples in the prompt so that the prompt concept is “selected” through marginalization.



Source: Xie et al., [An Explanation of In-context Learning as Implicit Bayesian Inference](#), July 2022

Prompts provide noisy evidence for Bayesian inference

- The transitions within training examples (green in the figure) allow the LM to infer the latent concept they all share. In a prompt, the green transitions come from below, which all provide signal for Bayesian inference.
 - the input distribution (the transitions inside the news sentence),
 - the output distribution (the topic word),
 - the format (syntax of news sentence),
 - the input-output mapping (relation between the news and the topic) Because the training examples are IID, concatenating them together often creates unnatural, low-probability transitions between examples (red in the figure) .
- if the signal is greater than the noise, then the LM can successfully do in-context learning:

Circulation revenue has increased by 5% in Finland. // Finance

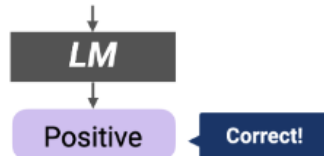
They defeated ... in the NFC Championship Game. // Sports

Apple ... development of in-house chips. // Tech

Which aspects of the prompt are most important for in-context learning?

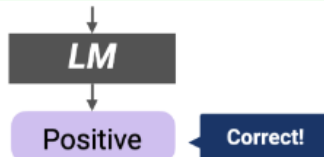
Examples with ground truth outputs

Circulation revenue has increased by 5% in Finland. \n Positive
Panostaja did not disclose the purchase price. \n Neutral
Paying off the national debt will be extremely painful. \n Negative
The company anticipated its operating profit to improve. \n _____



Examples with random outputs

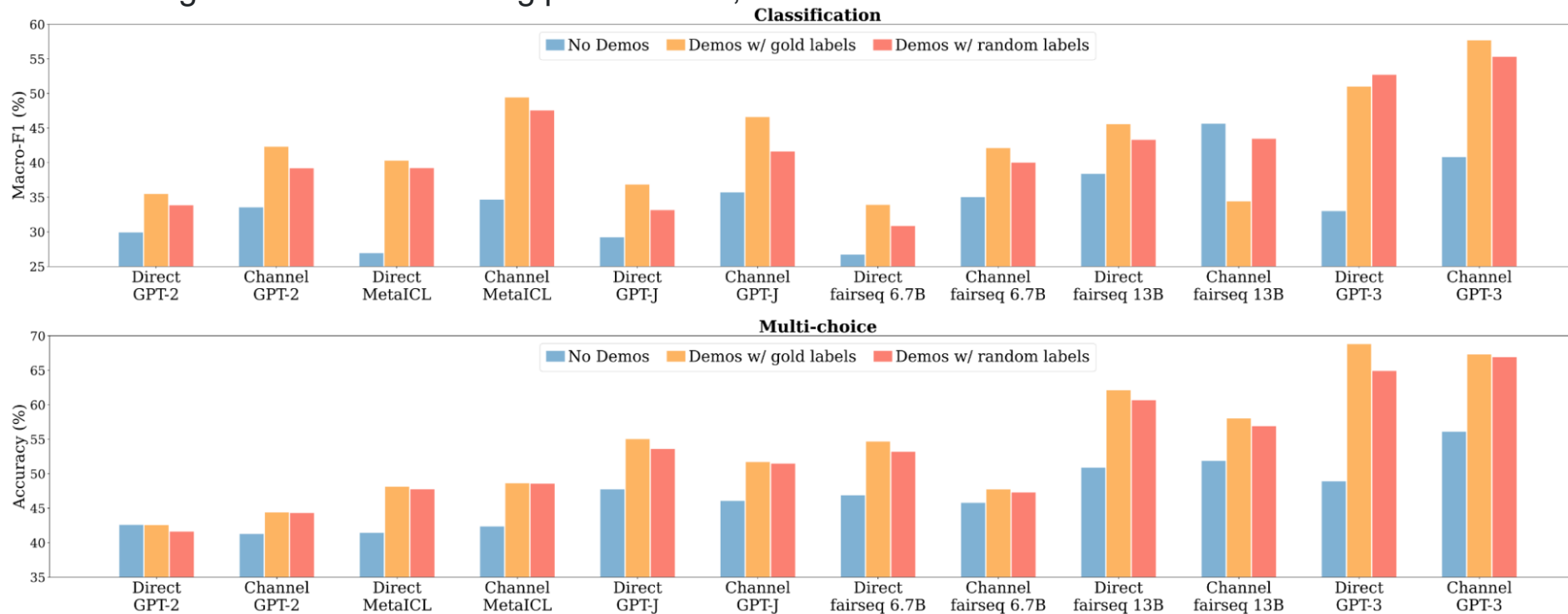
Circulation revenue has increased by 5% in Finland. \n **Neutral**
Panostaja did not disclose the purchase price. \n **Negative**
Paying off the national debt will be extremely painful. \n **Positive**
The company anticipated its operating profit to improve. \n _____



- Input-output pairing in the prompt matters much less than previously thought

The prompt with ground truth outputs (top) and the prompt with random outputs (bottom).

- Aphthous experiment with 12 models whose sizes range 774M - 175B, including the largest GPT-3.
- Replacing ground truth outputs with random outputs hurts performance significantly less than previously thought and is still significantly better than no-examples.
- This means, unlike typical supervised learning, ground truth outputs are not really required to achieve good in-context learning performance, which is counter-intuitive!



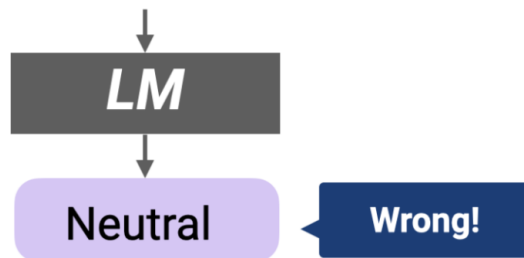
Comparison between no-examples (blue), examples with ground truth outputs (yellow) and examples with random outputs (random) 20

Which aspects of the prompt are most important for in-context learning? *Continued*

- **The input distribution matters:** when the inputs in the prompt are replaced with random inputs from an external corpus (the CC News corpus), model performance significantly drops. This indicates conditioning on the correct input distribution is important.

Colour-printed lithograph. Very good condition.	\n	Neutral
Many accompanying marketing ... meaning.	\n	Negative
In case you are interested in learning more about ...	\n	Positive
The company anticipated its operating profit to improve.	\n	_____

**Randomly Sampled from CC News*



Which aspects of the prompt are most important for in-context learning? *Continued*

- **The output space matters:** when the outputs in the examples are replaced with random English unigrams, model performance significantly drops. This indicates conditioning on the correct output space is important.

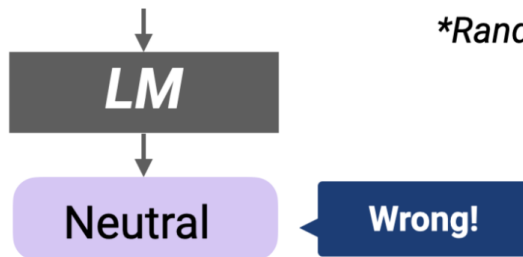
Circulation revenue has increased by 5% in Finland. \n Unanimity

Panostaja did not disclose the purchase price. \n Wave

Paying off the national debt will be extremely painful. \n Guana

The company anticipated its operating profit to improve. \n _____

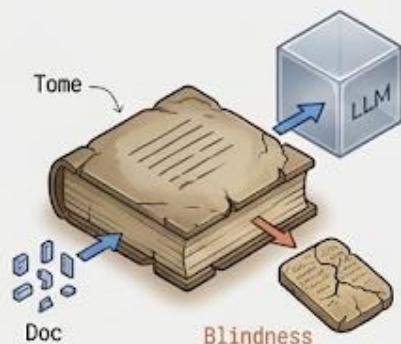
**Random English unigrams*



Insights

- We can theoretically view in-context learning as Bayesian inference of a latent concept conditioned on the prompt, and this capability comes from structure (long-term coherence) in the pretraining data.
 - in-context learning is mainly about locating latent concepts learned during pretraining.
 - If terms in a particular instance are exposed many times in the pretraining data, the model is likely to know better about the distribution of the inputs.
- The in-context learning still works well when outputs in the prompt are replaced with random outputs.
 - While random outputs add noise and remove input-output mapping information, other components (input distribution, output distribution, format) still provide evidence for Bayesian inference.
- Specifying the task description can improve Bayesian inference by providing explicit observations of the latent prompt concept.

From Static Retrieval to Introspection (2020–2024)

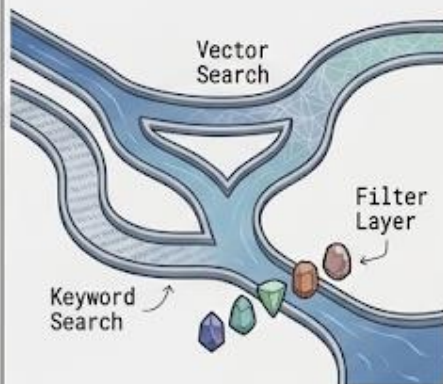


Naive RAG (The Baseline)

The Issue: Knowledge Cutoff. Models are blind to private data; retraining is costly.

The Innovation: Linear 'Search & Paste.' Vectorize docs, retrieve top-k, feed to LLM.

Critical Flaw: Blindness. Assumes retrieved docs are relevant; no quality verification.

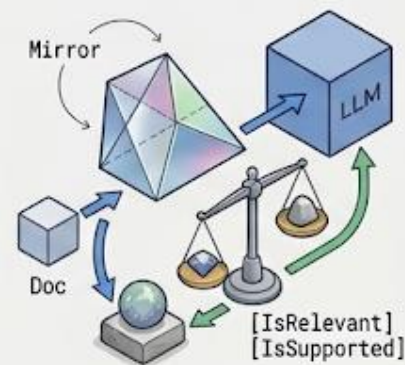


Advanced / Hybrid RAG

The Issue: 'Garbage in. Garbage Out.' Vague queries and missed keywords cause failure.

The Innovation: Pre & Post Optimization.

- **Pre-Retrieval:** Query Expansion & HyDE (Hypothetical Embeddings).
- **Retrieval:** Hybrid Search (Vector + BM25).
- **Post-Retrieval:** Cross-Encoder Reranking.

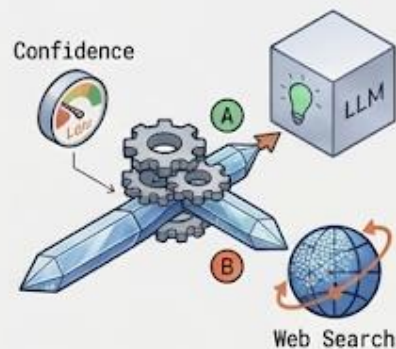


Self-RAG (The Critic)

The Issue: Blind Trust. Models hallucinate even with irrelevant docs.

The Innovation: Metacognition.

- **Reflection Tokens:** Outputs tags like [IsRelevant], [IsSupported].
- Critiques utility *before* answering.



CRAG (Corrective RAG)

The Issue: Retrieval Failure. Internal DB lacks the answer.

The Innovation: The 'Retrieval Evaluator.'

- Scores retrieved docs.
- **High Score:** Use docs.
- **Low Score:** Trigger Web Search fallback.

RAG vs. Fine-tune

